

LAB # 2 : Graphical Representation in *R*

Introduction

Graphical summaries are an effective way to communicate information, as visual displays are often easier to understand than numerical tables. The choice of a graphical method depends on the nature of the data and the objective of the analysis. In this lab, we explore various types of graphs in *R*. This lab is designed for one laboratory session.

After completing this lab, students will be able to create and interpret pie charts, dot plots, bar graphs, Pareto charts, histograms, box plots, and line plots using *R*.

Graphical Representation

- **Pie Chart**

A pie chart is a type of graph representing data in a circular form, with each slice of the circle representing a fraction or proportionate part of the whole. All slices of the pie add up to make the whole equaling 100 percent and 360 degrees.

For example, the following data is given about the number of students admitted in a University in Patiala.

City	Patiala	Chandigarh	Delhi	Other cities	International
Percentage of Students	42	18	15	13	12

```
x<- c(42, 18,15, 13,12)
labels <- c("Patiala", "Chandigarh", "Delhi", "Other Cities",
  "International")
pie(x, labels, main="Student Distribution by City")
```

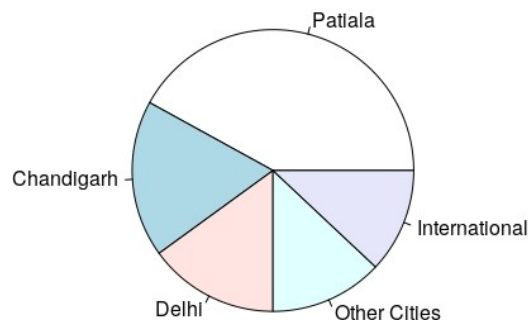


Figure 1: Pie Chart

- **Dot Plot**

Dot plots display numerical values along a number line. We can also make a dot plot for the same data given above.

```
x<- c(42, 18,15, 13,12)
labels <- c("Patiala", "Chandigarh", "Delhi", "Other Cities",
"International")
dotchart(x, labels,xlab="Percent")
```

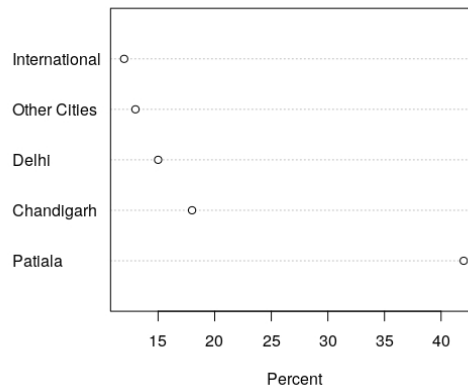


Figure 2: Dot Plot

- **Bar Graph**

Bar graphs are used to compare quantities across categories. The bar graph is essentially a dot plot put on its side.

For example, let us sketch the number of students with number of assignments completed.

```
A <- c(2,4,6,8)
B <- c(3, 4, 5, 6)
# Plot the bar chart
barplot(A, names.arg = B, xlab = "Number of Assignment Completed",
        ylab = "Number of Students", col = "green", main = "Bar Graph")
```

- **Pareto Chart**

Pareto chart is a combination of a bar chart and a line chart used for visualization where individual values are represented in descending order by bars, and the cumulative total is represented by the line. A package namely *qcc* is required.

A Pareto diagram can display of each type of failure or defect according to frequency, can help engineers identify important defects and their causes.

For example, the following data recorded from a power station:

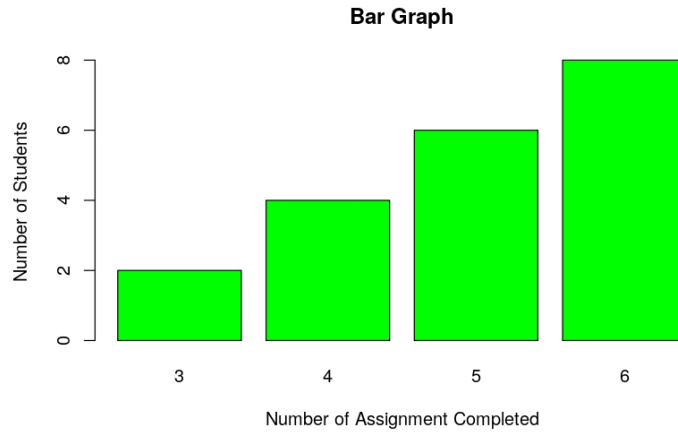


Figure 3: Bar Graph

Causes	Frequencies
power fluctuations	5
controller not stable	23
operator error	12
worn tool not replaced	3
other	4

```
install.packages('qcc')
library(qcc)
defect <- c(5,23,12,3,4)
names(defect) <- c("power fluctuations", "controller not stable",
  "operator error", "worn tool not replaced", "other")
pareto.chart(defect, xlab = "Defect", ylab="Count",
  col=heat.colors(length(defect)),
  cumperc = seq(0, 100, by = 20), ylab2 = "Cumulative Percentage",
  main = "Pareto Diagram")
```

Note: Pareto charts help identify the most significant causes contributing to problems.

- **Histogram**

Histograms represent the frequency distribution of continuous data. For frequency data, we can create a histogram by using *hist()* function.

```
302 315 265 296 289 301 308 305
280 285 318 267 300 309 312 314
299 316 301 286 281 311 272 318
295 305 283 309 313 278 284 308
296 291 310 302 282 287 307 280
```

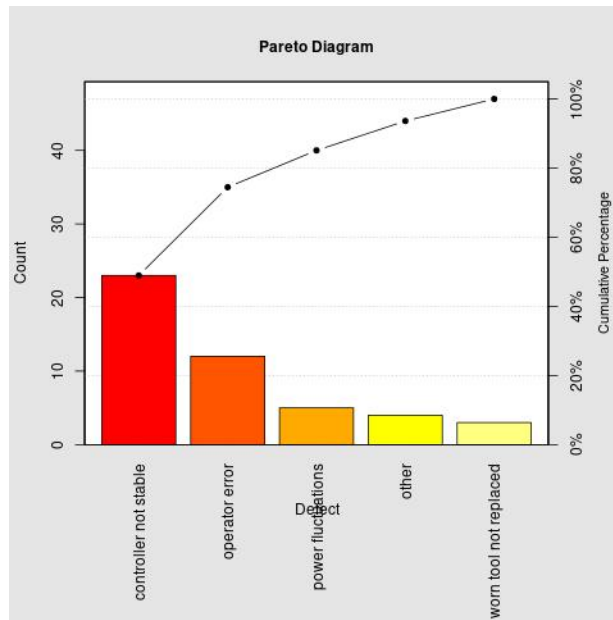


Figure 4: Pareto Diagram

```
x<- c(302, 315, 265, 296, 289, 301, 308, 305,
      280, 285, 318, 267, 300, 309, 312, 314,
      299, 316, 301, 286, 281, 311, 272, 318,
      295, 305, 283, 309, 313, 278, 284, 308,
      296, 291, 310, 302, 282, 287, 307, 280)
hist(x, col="lightblue",
     main="Histogram",
     xlab="Observed Values")
```

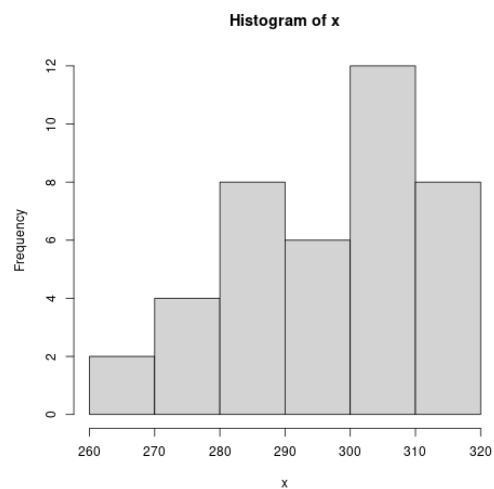
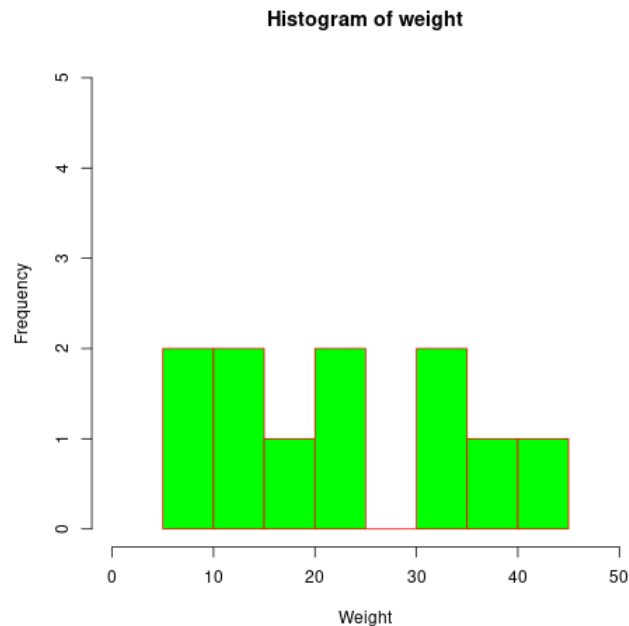


Figure 5: Histogram

We can also save the histogram file in specific format using labels. For example:

```
weight <- c(9,13,21,8,36,22,12,41,31,33,19)
hist(weight)
# if we want to save in specific format and using lables
png(file = "histogram_Weight.png")
# Create the histogram.
hist(weight, xlab = "Weight",col = "green",border = "red", xlim = c(0,50)
ylim = c(0,5), breaks = 5)
# Save the file.
dev.off()
```



• Box and Whisker Plot

Box plots display the distribution using quartiles and identify outliers. This plot is a convenient way of visually displaying the data distribution through their quartiles. We can make a box plot using *boxplot()* function.

For example, let us take a data of the marks of 10 students: 5, 39, 75, 79, 85, 90, 91, 93, 93, 98.

```
x<- c (5,39,75,79,85,90,91,93,93,98)
boxplot.default(x)
```

• Scatter Plot

A scatter plot is a type of plot used to display the relationship between two numerical variables, and plots one dot for each observations. For example, the following data is about the car age and its speed.

```
x <- c(5,7,8,7,2,2,9,4,11,12,9,6)
y <- c(99,86,87,88,111,103,87,94,78,77,85,86)
plot(x, y, main="Observation of Cars", xlab="Car Age",
```

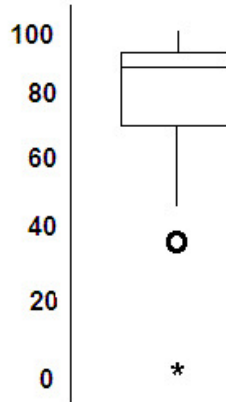


Figure 6: Box Plot with Outlier

`ylab="Car Speed")`

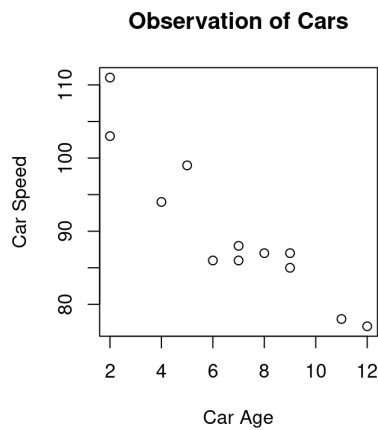


Figure 7: Scatter Plot

Problems:

1. In a study conducted at Virginia Tech, the steel rods supplied by two different companies were compared. Ten sample springs were made out of the steel rods supplied by each company, and a measure of flexibility was recorded for each. The data are as follows:

Company A:	9.3	8.8	6.8	8.7	8.5
	6.7	8.0	6.5	9.2	7.0
Company B:	11.0	9.8	9.9	10.2	10.1
	9.7	11.0	11.1	10.2	9.6

- (a) Calculate the sample mean and median for the data for the two companies.

- (b) Plot the data for the two companies on the same line and give your impression regarding any apparent differences between the two companies.
2. The following data represents the weights (in kg) of 30 engineering students:
- 52, 55, 58, 60, 61, 63, 65, 67, 68, 70, 72, 73, 75, 76, 78, 80, 82, 83, 85, 86, 88, 90, 92, 94, 95
- (a) Construct a histogram using R.
- (b) Comment on the shape of the distribution.
3. The marks obtained by two sections of students in a statistics test are given below:
- Section A: 45, 52, 60, 63, 68, 72, 75, 78, 81, 85
- Section B: 50, 55, 62, 66, 70, 74, 77, 80, 84, 88
- (a) Draw side-by-side box plots for the two sections.
- (b) Compare the median and variability of marks.
4. A manufacturing unit recorded the following number of defects during inspection:

Defect Type	Frequency
Surface defect	18
Crack	25
Misalignment	10
Improper finishing	7
Other	5

- (a) Construct a Pareto chart using R.
- (b) Identify the most significant defect contributing to quality loss.